

# Progress Report Sprint 5

- Course Name: CS461 Senior Software Engineering Project I
- Date: 12/04/25
- Team Name: TaxonBodyMassML
- Team ID: CS.032
- Student Names and Emails:
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## Summary

- Added content to the “Help” and “Resources” pages
- Updated JavaScript functionality
- Completed data merging
- Produced stakeholder presentation

## Progress vs Plan

| PLANNED ITEM  | STATUS (% OR DONE/PARTIAL/NOT DONE) | LINKS (ISSUE/PR)                                                                                                                                                                                        | NOTES/BLOCKERS                                                  | PLAN CHANGE                                                                             |
|---------------|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| HELP PAGE     | Partial                             | <a href="https://github.com/marknovak/TaxonBodyMassML/tree/web-dev/web_dev">https://github.com/marknovak/TaxonBodyMassML/tree/web_dev/web_dev</a>                                                       | Want input from external users before implementing more content |                                                                                         |
| DECISION TREE | Partial                             | <a href="https://github.com/marknovak/TaxonBodyMassML/commit/3b5f58fffe71f521b50bc4adb61dc2720e9a70d9">https://github.com/marknovak/TaxonBodyMassML/commit/3b5f58fffe71f521b50bc4adb61dc2720e9a70d9</a> | Data merge is complete, so we can now work on this portion.     |                                                                                         |
| DATA MERGING  | Done                                | <a href="https://github.com/marknovak/TaxonBodyMassML/tree/data-combination">https://github.com/marknovak/TaxonBodyMassML/tree/data-combination</a>                                                     |                                                                 | This section is complete, so we can redirect efforts towards the LLM predictive method. |

## Evidence of Working Software

Web Development latest update commit hash: 7761a515418212f98f861415a8657abafd51dd87

Link to UI: [https://praterh.github.io/HaileysTaxonBodyMassML/web\\_dev/index.html](https://praterh.github.io/HaileysTaxonBodyMassML/web_dev/index.html)

Videos of question submission and permanence (respectively):  [questions.webm](#)  [permanence.webm](#)

## Risk and Quality

- Some class triggers and movements in the JavaScript become interconnected.
  - Owner: Hailey
  - Status: In-Progress
  - Concrete actions: My goal is to make class toggles for modals less reliant on the toggle of others. I am going to change the logic of the JavaScript to rely on features of the page rather than actions. For example, I had to change the input box to toggle its “hidden” class if the intro modal’s class list contained “hidden”, not just when the intro modal was closed.
- Possible that dataset is too large for the decision tree to handle
  - Owner: Grant Pasquantonio
  - Status: In-Progress
  - Concrete Actions: Move straight to random forest implementation, which samples with replacement from the larger dataset. Alternatively, consider using a specialized library to compensate for the dataset size to avoid memory overflow.

## Next Goals

- Get more content for the “Help” page by getting input from people external from the project; these people should show a range of technical literacy. This effort will contribute to REQ-004 and REQ-006, including a help page and adhering to accessibility guidelines.
- Implement a fully functional prototype decision tree. This complies with REQ-008 because it will help us ensure that the technique will produce accurate results.
- Construct Newick string processing method to incorporate phylogenetic data into LLM prediction component. This will help us achieve REQ-001 because the LLM component will be able to generate an effective confidence interval for each prediction.

## Individual Contributions

Grant:

- Working on branches: <https://github.com/marknovak/TaxonBodyMassML/tree/random-forest>, <https://github.com/marknovak/TaxonBodyMassML/tree/data-partitioning>
- Organized and recorded the stakeholder presentation
- Sent video along w/ explanatory message to project partner
- Further preparation on decision tree prototype methods before data merge completion

Hailey:

- Working on branch: <https://github.com/marknovak/TaxonBodyMassML/tree/web-dev>
- Added content to “Resources” page
- Added content to “Help” page
- Tweaked JavaScript functionality to better suit “Help” page content
- Introduced question forum and permanence

Anish:

- Working on branch: <https://github.com/marknovak/TaxonBodyMassML/tree/data-combination> and <https://github.com/marknovak/TaxonBodyMassML/tree/test-cases>
- Finished data merge between taxonomic data and body mass data

- Cleaned some of the columns which do not include any taxonomic data
- Finding additional sources for species with no taxonomic data

## Team Reflection

Although the three of us were exceedingly busy this sprint, what with dead week and preparation for finals, we were happy with what we got done. There was good progress on the UI front, with content being added to the “Help” and “References” pages for the first time. Having content on these pages feels good because having both of these pages was included on our list of requirements. The layout of the UI also feels complete at this point, so we can jump into making it look good next term.

In regard to our algorithm efforts, we have finalized our plans for implementation. We will complete the test version of the random forest via the prototype decision tree. Then we can begin the random forest model implementation. Furthermore, we can also begin trying different variations of the LLM implementation using Newick strings to represent phylogenetic relationships.

Overall, we know we could have gotten more done this sprint, but we feel like we’re in a great place to hit the ground running next term.